

# Commerce XI Maths App

## Exercise 1.1

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 1 - Partition Values | Total questions: 18

### Question 1 - Enter answer · PG-7-8 · EX-Q1(i)

For series of observations : 16, 14.9, 11.5, 11.8, 11.1, 14.5, 14, 12, 10.9, 10.7, 10.6, 10.5, 13.5, 13, 12.6

Compute quartile  $Q_1$

**Correct answer: 10.9**

Explanation and steps

Arrange raw observations when required, locate  $i(n+1)/4$ , or use  $Q_i = L + (h/f)(iN/4 - c.f.)$  for grouped data. The result is 10.9.

### Question 2 - Enter answer · PG-7-8 · EX-Q1(ii)

For series of observations : 16, 14.9, 11.5, 11.8, 11.1, 14.5, 14, 12, 10.9, 10.7, 10.6, 10.5, 13.5, 13, 12.6

Compute quartile  $Q_2$

**Correct answer: 12**

Explanation and steps

Arrange raw observations when required, locate  $i(n+1)/4$ , or use  $Q_i = L + (h/f)(iN/4 - c.f.)$  for grouped data. The result is 12.

### Question 3 - Enter answer · PG-7-8 · EX-Q1(iii)

For series of observations : 16, 14.9, 11.5, 11.8, 11.1, 14.5, 14, 12, 10.9, 10.7, 10.6, 10.5, 13.5, 13, 12.6

Compute quartile  $Q_3$

**Correct answer: 14**

Explanation and steps

Arrange raw observations when required, locate  $i(n+1)/4$ , or use  $Q_i = L + (h/f)(iN/4 - c.f.)$  for grouped data. The result is 14.

### Question 4 - Enter answer · PG-7-8 · EX-Q2(i)

The heights (in cm) of 10 students are 148, 171, 158, 151, 154, 159, 152, 163, 171, 145.

Calculate  $Q_1$

**Correct answer: 150.25**

Explanation and steps

Arrange raw observations when required, locate  $i(n+1)/4$ , or use  $Q_i = L + (h/f)(iN/4 - c.f.)$  for grouped data. The result is 150.25.

### Question 5 - Enter answer · PG-7-8 · EX-Q2(ii)

The heights (in cm) of 10 students are 148, 171, 158, 151, 154, 159, 152, 163, 171, 145.

Calculate  $Q_3$

**Correct answer: 165**

Explanation and steps

Arrange raw observations when required, locate  $i(n+1)/4$ , or use  $Q_i = L + (h/f)(iN/4 - c.f.)$  for grouped data. The result is 165.

### Question 6 - Enter answer · PG-7-8 · EX-Q3

Monthly consumption of electricity (in units) of families in a certain locality is given below :

205, 201, 190, 188, 194, 172, 210, 225, 215, 232, 260, 230.

Calculate electricity consumption (in units) below which 25% of families lie.

**Correct answer: 191**

Explanation and steps

Arrange raw observations when required, locate  $i(n+1)/4$ , or use  $Q_i = L + (h/f)(iN/4 - c.f.)$  for grouped data. The result is 191.

### Question 7 - Enter answer · PG-7-8 · EX-Q4

For the following data of daily expenditure of families (in ₹),

compute the expenditure below which 75% of families include their expenditure.

| Daily Expenditure (in ₹) | 350 | 450 | 550 | 650 | 750 |
|--------------------------|-----|-----|-----|-----|-----|
| No. of families          | 16  | 19  | 24  | 28  | 13  |

**Correct answer: 650**

Explanation and steps

Arrange raw observations when required, locate  $i(n+1)/4$ , or use  $Q_i = L + (h/f)(iN/4 - c.f.)$  for grouped data. The result is 650.

### Question 8 - Enter answer · PG-7-8 · EX-Q5(i)

Calculate quartile  $Q_1$  for the following frequency distribution :

| No. of E-transactions per day | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7 |
|-------------------------------|----|----|----|----|----|----|----|---|
| No. of days                   | 10 | 35 | 45 | 95 | 64 | 32 | 10 | 9 |

**Correct answer: 2**

Explanation and steps

Arrange raw observations when required, locate  $i(n+1)/4$ , or use  $Q_i = L + (h/f)(iN/4 - c.f.)$  for grouped data. The result is 2.

### Question 9 - Enter answer · PG-7-8 · EX-Q5(ii)

Calculate quartile  $Q_2$  for the following frequency distribution :

| No. of E-transactions per day | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7 |
|-------------------------------|----|----|----|----|----|----|----|---|
| No. of days                   | 10 | 35 | 45 | 95 | 64 | 32 | 10 | 9 |

**Correct answer: 3**

Explanation and steps

Arrange raw observations when required, locate  $i(n+1)/4$ , or use  $Q_i = L + (h/f)(iN/4 - c.f.)$  for grouped data. The result is 3.

### Question 10 - Enter answer · PG-7–8 · EX-Q5(iii)

Calculate quartile  $Q_3$  for the following frequency distribution :

| No. of E-transactions per day | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7 |
|-------------------------------|----|----|----|----|----|----|----|---|
| No. of days                   | 10 | 35 | 45 | 95 | 64 | 32 | 10 | 9 |

**Correct answer: 4**

Explanation and steps

Arrange raw observations when required, locate  $i(n+1)/4$ , or use  $Q_i = L + (h/f)(iN/4 - c.f.)$  for grouped data. The result is 4.

### Question 11 - Enter answer · PG-7–8 · EX-Q6

The following is the frequency distribution of heights of 200 male adults in a factory :

| Height in cm. | No. of male adults |
|---------------|--------------------|
| 145–150       | 4                  |
| 150–155       | 6                  |
| 155–160       | 25                 |
| 160–165       | 57                 |
| 165–170       | 64                 |
| 170–175       | 30                 |
| 175–180       | 8                  |
| 180–185       | 6                  |

Find the central height.

**Correct answer: 165.625**

Explanation and steps

Arrange raw observations when required, locate  $i(n+1)/4$ , or use  $Q_i = L + (h/f)(iN/4 - c.f.)$  for grouped data. The result is 165.625.

### Question 12 - Enter answer · PG-7–8 · EX-Q7(i)

The following is the data of pocket expenditure per week of 50 students in a class.  
It is known that the median of the distribution is ₹120.

| Expenditure per week (in ₹) | 0–50 | 50–100 | 100–150 | 150–200 | 200–250 |
|-----------------------------|------|--------|---------|---------|---------|
| No. of students             | 7    | a      | 15      | b       | 3       |

Find missing frequency a.

**Correct answer: 12**

Explanation and steps

Arrange raw observations when required, locate  $i(n+1)/4$ , or use  $Q_i = L + (h/f)(iN/4 - c.f.)$  for grouped data. The result is 12.

### Question 13 - Enter answer · PG-7–8 · EX-Q7(ii)

The following is the data of pocket expenditure per week of 50 students in a class.  
It is known that the median of the distribution is ₹120.

| Expenditure per week (in ₹) | 0-50 | 50-100 | 100-150 | 150-200 | 200-250 |
|-----------------------------|------|--------|---------|---------|---------|
| No. of students             | 7    | a      | 15      | b       | 3       |

Find missing frequency b.

**Correct answer: 13**

Explanation and steps

Arrange raw observations when required, locate  $i(n+1)/4$ , or use  $Q_i = L + (h/f)(iN/4 - c.f.)$  for grouped data. The result is 13.

### Question 14 - Enter answer · PG-7–8 · EX-Q8(i)

The following is the distribution of 160 workers according to the wages in a certain factory :

| Wages more than (in ₹) | No. of workers |
|------------------------|----------------|
| 8000                   | 160            |
| 9000                   | 155            |
| 10000                  | 137            |
| 11000                  | 103            |
| 12000                  | 57             |
| 13000                  | 23             |
| 14000                  | 10             |
| 15000                  | 1              |
| 16000                  | 0              |

Determine quartile  $Q_1$ .

**Correct answer: 10500**

Explanation and steps

Arrange raw observations when required, locate  $i(n+1)/4$ , or use  $Q_i = L + (h/f)(iN/4 - c.f.)$  for grouped data. The result is 10500.

### Question 15 - Enter answer · PG-7–8 · EX-Q8(ii)

The following is the distribution of 160 workers according to the wages in a certain factory :

| Wages more than (in ₹) | No. of workers |
|------------------------|----------------|
| 8000                   | 160            |
| 9000                   | 155            |
| 10000                  | 137            |
| 11000                  | 103            |
| 12000                  | 57             |
| 13000                  | 23             |
| 14000                  | 10             |
| 15000                  | 1              |
| 16000                  | 0              |

Determine quartile  $Q_2$ .

**Correct answer: 11500**

Explanation and steps

Arrange raw observations when required, locate  $i(n+1)/4$ , or use  $Q_i = L + (h/f)(iN/4 - c.f.)$  for grouped data. The result is 11500.

### Question 16 - Enter answer · PG-7–8 · EX-Q8(iii)

The following is the distribution of 160 workers according to the wages in a certain factory :

| Wages more than (in ₹) | No. of workers |
|------------------------|----------------|
| 8000                   | 160            |
| 9000                   | 155            |
| 10000                  | 137            |
| 11000                  | 103            |
| 12000                  | 57             |
| 13000                  | 23             |
| 14000                  | 10             |
| 15000                  | 1              |
| 16000                  | 0              |

Determine quartile  $Q_3$ .

**Correct answer: 12500**

Explanation and steps

Arrange raw observations when required, locate  $i(n+1)/4$ , or use  $Q_i = L + (h/f)(iN/4 - c.f.)$  for grouped data. The result is 12500.

### Question 17 - MCQ · PG-7-8 · EX-Q9

Following is grouped data for duration of fixed deposits of 100 senior citizens from a certain bank :

| Fixed deposits (in days) | 0-180 | 180-360 | 360-540 | 540-720 | 720-900 |
|--------------------------|-------|---------|---------|---------|---------|
| No. of senior citizens   | 15    | 20      | 25      | 30      | 10      |

Calculate the limits of fixed deposits of central 50% senior citizens.

Options

- A. (270, 630)
- B. (270, 360)
- C. (180, 630)
- D. (360, 630)

**Correct answer: (270, 630)**

Explanation and steps

$N = 100$ .  $Q_1$  is the 25th item:  $180 + [(25-15)/20] \times 180 = 270$ .  $Q_3$  is the 75th item:  $540 + [(75-60)/30] \times 180 = 630$ . The printed answer 360 in the textbook key is an error.

### Question 18 - Enter answer · PG-7-8 · EX-Q10

Find the missing frequency given that the median of distribution is 1504.

| Life in hours | 950-1150 | 1150-1350 | 1350-1550 | 1550-1750 | 1750-1950 | 1950-2150 |
|---------------|----------|-----------|-----------|-----------|-----------|-----------|
| No. of bulbs  | 20       | 43        | 100       | x         | 23        | 13        |

**Correct answer: 81**

Explanation and steps

Arrange raw observations when required, locate  $i(n+1)/4$ , or use  $Q_i = L + (h/f)(iN/4 - c.f.)$  for grouped data. The result is 81.